Compute Moran Eigenvector Maps for Spatial Filtering

Description

Computes Moran Eigenvector Maps (MEMs) from projected core locations using the shared exact or low-rank strategy and returns the public predictor data frame used by [prepare_spatial_predictors_for_fit()].

Usage

```
compute_spatial_mev(
  data_coords_projected = NULL,
  n_mev = 20L,
  strategy = "exact",
  exact_max_locations = 1999L,
  fast_eigenvectors = 200L,
  fast_seed = 900723L
)
```

Arguments

`data_coords_projected`	A data frame with 'dataset_name' as row names and columns 'coord_x_km' and 'coord_y_km', as returned by 'project_coords_to_metric()'. Must have more than 3 rows (required by 'sjSDM::generateSpatialEV()'). Each row represents one core/site location.
`n_mev`	A positive integer giving the number of eigenvectors to return. If 'n_mev' exceeds the available count, it is clamped down with a warning. Default is '20L'.
`strategy`	Shared construction strategy: '"exact"', '"fast"', or '"auto"'. Automatic selection uses the shared location boundary.
`exact_max_locations`	Largest input allowed to use dense exact construction. Default is '1999L'.
`fast_eigenvectors`	Low-rank basis size reserved for fast construction. It must be at least 'n_mev'. Default is '200L'.

`fast_seed`	Positive integer used locally for deterministic fast-basis knot selection. Default is '900723L'.

Details

MEMs capture the spatial autocorrelation structure of the sampling locations and are used as spatial predictors in the sjSDM model. Eigenvectors are computed on the unique core locations; the caller is responsible for expanding to sample level via 'prepare_spatial_predictors_for_fit()'.

Exact construction uses 'sjSDM::generateSpatialEV()'. Fast construction uses the package-backed Nyström basis from 'spmoran::meigen_f()'. This compatibility wrapper returns only the public values; callers that need held-out projection state or provenance should use [compute_spatial_mev_basis()].

Value

A data frame with the same row names as 'data_coords_projected', and 'n_mev' columns named 'mev_1', 'mev_2', …, 'mev_n_mev', containing the first 'n_mev' Moran eigenvectors. This data frame is a drop-in replacement for 'dplyr::select(data_coords_projected, coord_x_km, coord_y_km)' as input to 'prepare_spatial_predictors_for_fit()'.

See Also

[project_coords_to_metric()], [prepare_spatial_predictors_for_fit()], [scale_spatial_for_fit()]